

TOPIC 46/60

What are Vector Databases?

The Backbone of Modern AI.

How we store, index, and query high-dimensional semantic meaning at production scale.

Ashish Jangra

linkedin.com/in/ashish-jangra



THE CONTEXT GAP

SQL is Conceptually Blind

Relational databases are structured for exact matches—integers, dates, and strings. They are mathematically incapable of understanding that "kitten" and "feline" are conceptually the same.

The Keyword Failure

Traditional databases search for direct keyword intersects. If you search for "automobile repairs" and your document reads "car maintenance", a standard SQL engine finds exactly zero matches.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE DEFINITION

Spatial Memory



What is a Vector Database?

A vector database is a database specifically engineered to store, manage, and search high-dimensional vector representations (embeddings).

Instead of searching through rows and columns, a Vector DB treats query strings as points in space and finds documents that occupy the closest coordinates.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE SCALING PROBLEM

The Speed Bottleneck

Comparing a query vector against millions of document vectors one-by-one is known as **K-Nearest Neighbors (KNN)**. It scales at $O(N)$ speed—which is completely unusable in production.

Approximate Nearest Neighbor (ANN)

To solve the $O(N)$ lag, Vector DBs use ANN algorithms. They sacrifice a minuscule amount of mathematical accuracy in exchange for logarithmic $O(\log N)$ search speeds.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



ALGORITHM 1

The HNSW Index

Hierarchical Navigable Small World (HNSW) is the gold-standard graph algorithm for indexing vector space.

// The "Express Highway" Analogy

Top Layers (Fast lanes): Long-distance jumps spanning completely different topics.

Middle Layers (Highways): Medium-range navigation within broad category nodes.

Bottom Layer (Local streets): Ultra-dense connections linking micro-variants of specific concepts.

** The algorithm leaps across macro-clusters first, descending layers to locate local matches in microseconds.*

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



COMPRESSION

Clustering & Quantization

Vector arrays occupy massive RAM footprints. Vector DBs use two-pronged compression to scale down cost:

Inverted File Index (IVF)

Partitions the coordinate space into Voronoi cells. During search, you skip 99% of the cells and only compute similarities within the closest cluster.

Product Quantization (PQ)

Breaks massive vector dimensions into small segments and maps floating points to low-bit integers, shrinking RAM requirements by up to 95%.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



KEY FEATURE

Metadata Filtering

Vector DBs aren't just similarity engines; they are dual structured engines that handle metadata alongside spatial embeddings.

"Find passages resembling 'bypass valve guidelines' BUT ONLY if: [file_type: PDF] AND [region: EMEA]"

Vector databases perform **Single-Stage Filtering**, evaluating metadata parameters and vector vectors simultaneously to avoid the inefficiencies of post-filtering.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



SEARCH CONVERGENCE

Hybrid Search (RRF)

For maximum reliability, enterprise architectures blend both dense vectors and sparse keywords using Reciprocal Rank Fusion:

Dense Vector

Captures conceptual nuances and abstract phrasing (Semantic matches).

+

Sparse BM25

Catches strict part IDs, system model names, and exact serial codes.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



ARCHITECTURE SELECTION

The Landscape Matrix

CLOUD
NATIVE

Pinecone, Zilliz. Fully managed, serverless scaling, low operation overhead. Ideal for lean cloud startups.

SELF-HOSTED
/ OPEN

Qdrant, Milvus, Weaviate. Extensible, open-source architectures. High customizability for bare-metal or Kubernetes setups.

RELATIONAL
EXTENSIONS

pgvector (Postgres). Perfect if you already run Postgres. It keeps all transactional metadata and vector indices in a single relational home.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



Vector Databases

CONCEPT UNLOCKED



Computes similarity scores using spatial geometry



Navigates vectors in sub-linear time via HNSW graph mapping



Handles structured metadata filter requirements



Leverages IVF-PQ quantization to compress footprint

CONNECT WITH ME



LinkedIn



GitHub



Kaggle



Instagram

Ashish Jangra

linkedin.com/in/ashish-jangra

